

141. If $f(x) = \begin{vmatrix} x^3 & 2x^2+1 & 1+3x \\ 3x^2+2 & 2x & x^3+6 \\ x^3-x & 4 & x^2-2 \end{vmatrix}$ for all $x \in \mathbb{R}$, then $2f(0) + f'(0)$ is equal to :

यदि सभी $x \in \mathbb{R}$ के लिए $f(x) = \begin{vmatrix} x^3 & 2x^2+1 & 1+3x \\ 3x^2+2 & 2x & x^3+6 \\ x^3-x & 4 & x^2-2 \end{vmatrix}$ हैं, तो $2f(0) + f'(0)$ बराबर है :

(A) 48

(B) 18

(C) 24

(D) 42

Ans. D

Sol. $f(x) = \begin{vmatrix} x^3 & 2x^2+1 & 1+3x \\ 3x^2+2 & 2x & x^3+6 \\ x^3-x & 4 & x^2-2 \end{vmatrix}$

$$f'(x) = \begin{vmatrix} 3x^2 & 4x & 3 \\ 3x^2+2 & 2x & x^3+6 \\ x^3-x & 4 & x^2-2 \end{vmatrix} + \begin{vmatrix} x^3 & 2x^2+1 & 1+3x \\ 6x & 2 & 3x^2 \\ x^3-x & 4 & x^2-2 \end{vmatrix} + \begin{vmatrix} x^3 & 2x^2+1 & 1+3x \\ 3x^2+2 & 2x & x^3+6 \\ 3x^2-1 & 0 & 2x \end{vmatrix}$$

$$2f(0) + f'(0) = 2 \begin{vmatrix} 0 & 1 & 1 \\ 2 & 0 & 6 \\ 0 & 4 & -2 \end{vmatrix} + \begin{vmatrix} 0 & 0 & 3 \\ 2 & 0 & 6 \\ 0 & 4 & -2 \end{vmatrix} + \begin{vmatrix} 0 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 4 & -2 \end{vmatrix} + \begin{vmatrix} 0 & 1 & 1 \\ 2 & 0 & 6 \\ -1 & 0 & 0 \end{vmatrix}$$

$$= 24 + 24 + 0 - 6$$

$$= 42$$

142. Let $g(x)$ be a linear function and $f(x) = \begin{cases} g(x) & , x \leq 0 \\ \left(\frac{1+x}{2+x}\right)^{\frac{1}{x}} & , x > 0 \end{cases}$, is continuous at $x=0$. If $f'(1) = f(-1)$, then the

value of $g(3)$ is :

माना $g(x)$ एक रैखिक फलन है तथा $f(x) = \begin{cases} g(x) & , x \leq 0 \\ \left(\frac{1+x}{2+x}\right)^{\frac{1}{x}} & , x > 0 \end{cases}$, $x=0$ पर संतत है। यदि $f'(1) = f(-1)$ है, तो $g(3)$ का

मान है :

(A) $\log_e \left(\frac{4}{9e^3} \right)$ (B) $\log_e \left(\frac{4}{9} \right) - 1$ (C) $\frac{1}{3} \log_e \left(\frac{4}{9} \right) + 1$ (D) $\frac{1}{3} \log_e \left(\frac{4}{9e^3} \right)$

Ans. A

Sol. Let $g(x) = ax + b$

$f(x)$ is continuous at $x = 0$

$$\therefore \lim_{x \rightarrow 0^+} \left(\frac{1+x}{2+x} \right)^{\frac{1}{x}} = \left(\frac{1}{2} \right)^{\infty} = 0$$

$$\lim_{x \rightarrow 0^-} g(x) = \lim_{x \rightarrow 0^-} ax + b = b$$

So, $b = 0$

$$\text{Now, } f(x) = \left(\frac{1+x}{2+x} \right)^{\frac{1}{x}}$$

$$\ln f(x) = \frac{1}{x} [\ln(1+x) - \ln(2+x)]$$

$$\frac{1}{f(x)} f'(x) = \frac{x \left[\frac{1}{1+x} - \frac{1}{2+x} \right] - \ln \left(\frac{1+x}{2+x} \right)}{x^2}$$

$$f'(1) = f(1) \left[\frac{\left(\frac{1}{2} - \frac{1}{3} \right) - \ln \left(\frac{2}{3} \right)}{1} \right]$$

$$f'(1) = \frac{2}{3} \left[\frac{1}{6} - \ln \frac{2}{3} \right] \quad \dots(1)$$

Also $f(-1) = a(-1) + b$

$$f(-1) = -a \quad \dots(2)$$

$\therefore$ from (1) and (2)

$$-a = \frac{1}{9} - \frac{2}{3} \ln \frac{2}{3}$$

$$a = \frac{2}{3} \ln \left(\frac{2}{3} \right) - \frac{1}{9}$$

$\therefore g(x) = ax$

$$g(3) = 3 \left(\frac{2}{3} \ln \left(\frac{2}{3} \right) - \frac{1}{9} \right)$$

$$g(3) = 2 \ln \left(\frac{2}{3} \right) - \frac{1}{3}$$

$$g(3) = \ln \frac{4}{9} - \frac{1}{3}$$

$$g(3) = \ln \frac{4}{9} - \ln e^{\frac{1}{3}}$$

$$g(3) = \ln \left(\frac{4}{9e^{\frac{1}{3}}} \right)$$

$$g(3) = \log_e \left(\frac{4}{9e^{\frac{1}{3}}} \right)$$

143. Let $f(x) = \frac{\sin x + \cos x - \sqrt{2}}{\sin x - \cos x}$, $x \in [0, \pi] - \left\{ \frac{\pi}{4} \right\}$. Then $f\left(\frac{7\pi}{12}\right)f''\left(\frac{7\pi}{12}\right)$ is equal to :

यदि $f(x) = \frac{\sin x + \cos x - \sqrt{2}}{\sin x - \cos x}$, $x \in [0, \pi] - \left\{ \frac{\pi}{4} \right\}$ है, तो $f\left(\frac{7\pi}{12}\right)f''\left(\frac{7\pi}{12}\right)$ बराबर है :

- (A) $\frac{2}{9}$ (B) $-\frac{2}{3}$ (C) $\frac{2}{3\sqrt{3}}$ (D) $-\frac{1}{3\sqrt{3}}$

Ans. A

Sol.
$$f(x) = \sqrt{2} \frac{\left(\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x \right) - 1}{\sqrt{2} \left(\frac{1}{\sqrt{2}} \sin x - \frac{1}{\sqrt{2}} \cos x \right)} = \frac{\sin(x + \pi/4) - 1}{\sin(x - \pi/4)}$$

$$f'(x) = \frac{\cos(x + \pi/4) \sin(x - \pi/4) - \cos(x - \pi/4)(\sin(x + \pi/4) - 1)}{\sin^2(x - \pi/4)}$$

$$f'(x) = \frac{\cos(x - \pi/4) - 1}{\sin^2(x - \pi/4)} = \frac{-(1 - \cos(x - \pi/4))}{1 - \cos^2(x - \pi/4)}$$

$$f'(x) = -\frac{1}{1 + \cos(x - \pi/4)}$$

$$= f''(x) = -\frac{\sin(x - \pi/4)}{(1 + \cos(x - \pi/4))^2}$$

$$f(7\pi/12) = -\frac{1}{\sqrt{3}}$$

$$f''(7\pi/12) = -\frac{2\sqrt{3}}{9}$$

$$f(7\pi/12)f''(7\pi/12) = \frac{2}{9}$$

144. The value of $\lim_{x \rightarrow 0} 2 \left(\frac{1 - \cos x \sqrt{\cos 2x} \sqrt[3]{\cos 3x} \dots \sqrt[10]{\cos 10x}}{x^2} \right)$ is _____.

$$\lim_{x \rightarrow 0} 2 \left(\frac{1 - \cos x \sqrt{\cos 2x} \sqrt[3]{\cos 3x} \dots \sqrt[10]{\cos 10x}}{x^2} \right) \text{ का मान होगा } \underline{\hspace{2cm}}$$

Ans. 55

Sol. $\lim_{x \rightarrow 0} \frac{2}{x^2} \left(1 - \left(1 - \frac{x^2}{2} \right) \left(1 - \frac{2^2 x^2}{2} \right)^{\frac{1}{2}} \left(1 - \frac{3^2 x^2}{2} \right)^{\frac{1}{3}} \dots \left(1 - \frac{10^2 x^2}{2} \right)^{\frac{1}{10}} \right)$

$$\lim_{x \rightarrow 0} \frac{2}{x^2} \left(1 - \left(1 - \frac{x^2}{2} \right) \left(1 - \frac{2^2 x^2}{2.2} \right) \left(1 - \frac{3^2 \cdot x^2}{3.2} \right) \dots \left(1 - \frac{10^2 x^2}{2.10} \right) \right)$$

$$= \lim_{x \rightarrow 0} \frac{2}{x^2} \left(1 - \left(1 - \frac{x^2}{2} - \frac{2x^2}{2} - \frac{3x^2}{2} \dots - \frac{10x^2}{2} \right) \right)$$

$$= \left(\frac{2}{2} (1 + 2 + 3 + \dots + 10) \right) = \frac{10(10+1)}{2} = 55$$

145. Let $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$, $x \in \mathbb{R}$. Then $f'(10)$ is equal to _____.

माना $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$, $x \in \mathbb{R}$ है। तो $f'(10)$ बराबर है _____।

Ans. 202

Sol. $f(x) = x^3 + x^2 f'(1) + x f''(2) + f'''(3)$

$$f'(x) = 3x^2 + 2x f'(1) + f''(2)$$

$$f''(x) = 6x + 2f'(1)$$

$$f'''(x) = 6$$

$$\Rightarrow f'''(3) = 6$$

$$f''(2) = 12 + 2f'(1) \quad \text{_____ (1)}$$

$$f'(1) = 3 + 2f'(1) + f''(2) \quad \text{_____ (2)}$$

adding (1) and (2)

$$0 = 15 + 3f'(x) \Rightarrow f'(1) = -5$$

$$f''(2) = 12 - 10 = 2$$

Now

$$f'(10) = 300 + 20(-5) + 2$$

$$= 300 - 100 + 2 = 202$$

146. If $y = \frac{(\sqrt{x}+1)(x^2-\sqrt{x})}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}(3\cos^2 x - 5)\cos^3 x$, then $96y'\left(\frac{\pi}{6}\right)$ is equal to _____.

यदि $y = \frac{(\sqrt{x}+1)(x^2-\sqrt{x})}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}(3\cos^2 x - 5)\cos^3 x$ है, तो $96y'\left(\frac{\pi}{6}\right)$ बराबर है _____

Ans. 105

Sol. $y = \frac{(\sqrt{x}+1)(x^2-\sqrt{x})}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}(3\cos^2 x - 5)\cos^3 x$

$$y = \frac{(\sqrt{x}+1) \cdot \sqrt{x}(x\sqrt{x}-1)}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}(3\cos^2 x - 5)\cos^3 x$$

$$y = \frac{(\sqrt{x}+1) \cdot \sqrt{x}(\sqrt{x})^3 - 1^3}{x\sqrt{x}+x+\sqrt{x}} + \frac{1}{15}[3\cos^5 x - 5\cos^3 x]$$

$$y = \frac{(\sqrt{x}+1) \cdot \sqrt{x}(\sqrt{x}-1)(x+\sqrt{x}+1)}{\sqrt{x}(x+\sqrt{x}+1)} + \frac{1}{15}[3\cos^5 x - 5\cos^3 x]$$

$$y = (x-1) + \frac{1}{15}[3\cos^5 x - 5\cos^3 x]$$

Now, diff w.r.t x

$$y' = 1 - 0 + \frac{1}{15}[15\cos^4 x \cdot (-\sin x) - 15\cos^2 x(-\sin x)]$$

$$y' = 1 - \sin x \cdot [\cos^4 x - \cos^2 x]$$

$$y'\left(\frac{\pi}{6}\right) = 1 - \sin \frac{\pi}{6} \left[\cos^4 \frac{\pi}{6} - \cos^2 \frac{\pi}{6} \right]$$

$$= 1 - \frac{1}{2} \left[\frac{9}{16} - \frac{3}{4} \right]$$

$$= 1 - \frac{1}{2} \left[\frac{9-12}{16} \right] \Rightarrow 1 + \frac{3}{32} \Rightarrow \frac{35}{32}$$

$$96 \cdot y' \left(\frac{\pi}{6} \right) = 96 \times \frac{35}{32} \Rightarrow 105$$

Tangent & Normal

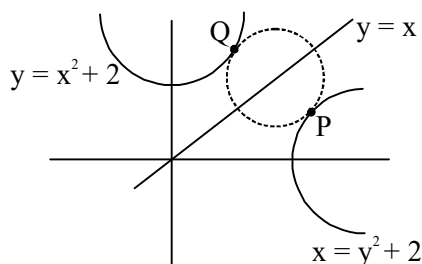
147. The radius of the smallest circle which touches the parabolas $y = x^2 + 2$ and $x = y^2 + 2$ is :

परवलयों $y = x^2 + 2$ और $x = y^2 + 2$ को स्पर्श करने वाले सबसे छोटे वृत्त की त्रिज्या है :

- (A) $\frac{7\sqrt{2}}{2}$ (B) $\frac{7\sqrt{2}}{8}$ (C) $\frac{7\sqrt{2}}{4}$ (D) $\frac{7\sqrt{2}}{16}$

Ans. B

Sol.



Tangent of P and Q must be parallel to $y = x$

$$\frac{dy}{dx} = 2x = 1$$

$$x = \frac{1}{2} \quad y = \frac{9}{4} \quad P \left(\frac{9}{4}, \frac{1}{2} \right) \quad Q \left(\frac{1}{2}, \frac{9}{4} \right)$$

$$PQ = \frac{7\sqrt{2}}{4} \quad r = \frac{7\sqrt{2}}{8}$$

148. A spherical chocolate ball has a layer of ice-cream of uniform thickness around it. When the thickness of the ice-cream layer is 1 cm, the ice-cream melts at the rate of $81 \text{ cm}^3/\text{min}$ and the thickness of the ice-cream layer decreases at the rate of $\frac{1}{4\pi} \text{ cm/min}$. The surface area (in cm^2) of the chocolate ball (without the ice-cream layer) is :

एक गोलाकार चॉकलेट बॉल पर इसके परितः एकसमान मोटाई की आइस-क्रीम की एक परत है। जब आइस-क्रीम की परत की मोटाई

1 cm है, तो आइसक्रीम $81 \text{ cm}^3/\text{min}$ की दर से पिघलती है और आइसक्रीम की परत की मोटाई $\frac{1}{4\pi} \text{ cm/min}$ की दर से घटती है।

चॉकलेट बॉल का (बिना आइसक्रीम की परत के) पृष्ठ क्षेत्रफल (cm^2 में) है :

- (A) 196π (B) 225π (C) 128π (D) 256π

Ans. D

Sol. Let thickness of ice cream be ' α ' cm

Initially $\alpha = 1 \text{ cm}$

$$\text{and } \frac{dv}{dt} = -81 \frac{\text{cm}^3}{\text{min}}$$

$$\text{Now } V = \frac{4}{3} \pi (r + \alpha)^3$$

$$\frac{dv}{dt} = \frac{4}{3} \pi \cdot 3(r + \alpha)^2 \cdot \frac{d\alpha}{dt} = \frac{4}{3} \pi \cdot 3(r + 1)^2 \times \frac{1}{4\pi}$$

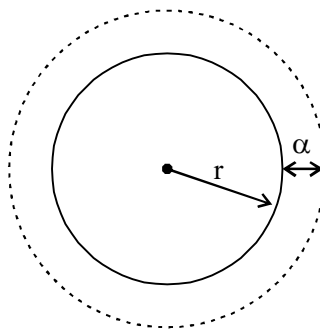
$$r + 1 = 9$$

$$r = 8$$

$$S = 4\pi \cdot r^2$$

$$= 4\pi \cdot 8^2$$

$$= 256\pi$$



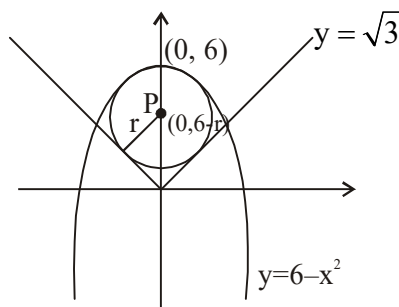
149. Let C be the circle of minimum area touching the parabola $y = 6 - x^2$ and the lines $y = \sqrt{3}|x|$. Then, which one of the following points lies on the circle C?

माना C न्यूनतम क्षेत्रफल का कोई वृत्त है जो परवलय $y = 6 - x^2$ और रेखा $y = \sqrt{3}|x|$ को स्पर्श कर रहा है। तब निम्नलिखित में से कौनसा बिन्दु वृत्त C पर स्थित होगा?

- (A) (1, 2) (B) (2, 2) (C) (1, 1) (D) (2, 4)

Ans. D

Sol.



$$\text{Equation of circle } x^2 + (y - 6 + r)^2 = r^2$$

distance between P and line = r

$$\left| \frac{6-r}{2} \right| = r$$

$$r = 2$$

Equation is $x^2 + (y-4)^2 = 4$ Point (2, 4) is lie on circle

150. The number of points on the curve $y = 54x^5 - 135x^4 - 70x^3 + 180x^2 + 210x$ at which the normal lines are parallel to $x + 90y + 2 = 0$ is :

वक्र $y = 54x^5 - 135x^4 - 70x^3 + 180x^2 + 210x$ पर उन बिंदुओं, जिन पर अभिलंब, रेखा $x + 90y + 2 = 0$ के समांतर है, की संख्या है :

- (A) 3 (B) 0 (C) 4 (D) 2

Ans. C

Sol. Normal of the line is parallel to line $x + 90y + 2 = 0$

$$m_N = -\frac{1}{90}$$

$$-\left(\frac{dx}{dy}\right)_{(x_1, y_1)} = -\frac{1}{90} \Rightarrow \left(\frac{dy}{dx}\right)_{(x_1, y_1)} = 90$$

Now,

$$\frac{dy}{dx} = 270x^4 - 540x^3 - 210x^2 + 360x + 210 = 90$$

$$\Rightarrow x = 1, 2, \frac{-2}{3}, \frac{-1}{3}$$

(D) normals

151. Let maximum and minimum values of $\left(\sqrt{8x - x^2 - 12} - 4\right)^2 + (x-7)^2$, $x \in \mathbb{R}$ be M and m, respectively. Then $M^2 - m^2$ is equal to _____.

माना $\left(\sqrt{8x - x^2 - 12} - 4\right)^2 + (x-7)^2$, $x \in \mathbb{R}$ के उच्चतम तथा निम्नतम मान क्रमशः M तथा m हैं। तो $M^2 - m^2$ बराबर है

Ans. 1600

Sol. $\left(\sqrt{8x - x^2 - 12} - 4\right)^2 + (x-7)^2 = (y-4)^2 + (x-7)^2$

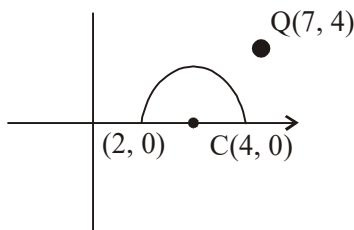
Where $y = \sqrt{8x - x^2 - 12} > 0$

$$\Rightarrow x^2 + y^2 - 8x + 12 = 0$$

$$\Rightarrow (x-4)^2 + y^2 = 4$$

Hence, $\left(\sqrt{8x - x^2 - 12} - 4\right)^2 + (x - 7)^2 = (PQ)^2$

Where P (x, y) lies on circle $y = \sqrt{8x - x^2 - 12}$ and Q (7, 4)



Now, minimum value of $(PQ)^2$ is :

$$m = (CQ - 2)^2 \text{ where C is centre}$$

$$= (5 - 2)^2 = 9$$

and maximum value of $(PQ)^2$ is :

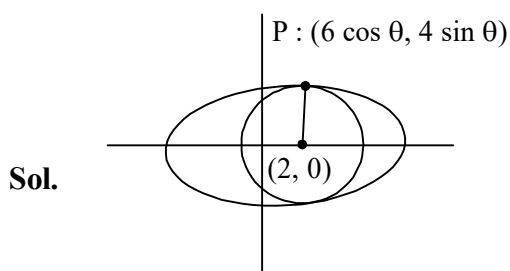
$$M = (7 - 2)^2 + (4 - 0)^2 = 25 + 16 = 41$$

$$\Rightarrow M^2 - m^2 = (41)^2 - (9)^2 = 50 \times 32 = 1600$$

152. Let C be the largest circle centred at (2, 0) and inscribed in the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$. If $(1, \alpha)$ lies on C, then $10\alpha^2$ is equal to _____.

माना C सबसे बड़ा वृत्त है जिसका केन्द्र (2, 0) पर है तथा जो दीर्घवृत्त $\frac{x^2}{36} + \frac{y^2}{16} = 1$ के अन्तर्गत है। यदि बिंदु $(1, \alpha)$ वृत्त C पर है, तो $10\alpha^2$ बराबर है _____

Ans. 118



Sol.

equation of ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$

$$\frac{x}{18} + \frac{yy'}{8} = 0$$

$$y' = \frac{-x}{18} \cdot \frac{8}{y}$$

$$Y_N = \frac{9y}{4x} = 9 \cdot \frac{4 \sin \theta}{4 \cdot 6 \cos \theta} = \frac{3 \sin \theta}{2 \cos \theta}$$

$$Y_N (\text{Slope}) = \frac{4 \sin \theta}{6 \cos \theta - 2} = \frac{2 \sin \theta}{3 \cos \theta - 1}$$

$$\frac{3 \sin \theta}{2 \cos \theta} = \frac{2 \sin \theta}{3 \cos \theta - 1}$$

$$\cos \theta = \frac{3}{5}$$

$$P: \left(\frac{18}{5}, \frac{16}{5} \right)$$

$$S = \sqrt{\left(\frac{8}{5} \right)^2 + \left(\frac{16}{5} \right)^2}$$

$$\text{Now } (x-2)^2 + y^2 = r^2$$

$$(1, \alpha) \rightarrow 1 + \alpha^2 = \frac{8^2}{5^2} + \frac{16^2}{5^2} = 12.8$$

$$\alpha^2 = 11.8$$

$$10\alpha^2 = 118$$

Monotonicity

153. Let f be a differentiable function on $\mathbb{R}$ such that $f(2) = 1$, $f'(2) = 4$. Let $\lim_{x \rightarrow 0} (f(2+x))^{3/x} = e^\alpha$.

Then the number of times the curve $y = 4x^3 - 4x^2 - 4(\alpha - 7)x - \alpha$ meets x -axis is :

माना $\mathbb{R}$ पर अवकलनीय एक फलन f के लिए $f(2) = 1$, $f'(2) = 4$ हैं। माना $\lim_{x \rightarrow 0} (f(2+x))^{3/x} = e^\alpha$ है। तो

वक्र $y = 4x^3 - 4x^2 - 4(\alpha - 7)x - \alpha$, x -अक्ष को कितनी बार मिलता है :

(A) 2

(B) 1

(C) 3

(D) 0

Ans. A

Sol. $\lim_{x \rightarrow 0} (f(2+x))^{3/x} = e^\alpha \quad (1^\infty)$

$$e^{\lim_{x \rightarrow 0} \frac{(f(2+x)-1) \cdot 3}{x}} = e^\alpha$$

$$e^{\lim_{x \rightarrow 0} \frac{f'(2+x)(3)}{1}} = e^\alpha$$

$$\Rightarrow \alpha = 12$$

$$\therefore y = 4x^3 - 4x^2 - 20x - 12$$

$$= 4(x^3 - x^2 - 5x - 3)$$

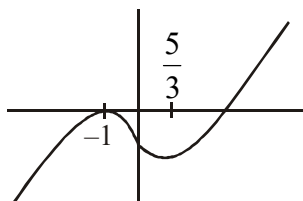
$$\frac{dy}{dx} = 4(3x^2 - 2x - 5) = 0$$

$$3x^2 - 5x + 3x - 5 = 0$$

$$(x + 1)(3x - 5) = 0$$

$$x = -1, \frac{5}{3}$$

$$y(-1) = 0; y\left(\frac{5}{3}\right) = 4\left(\frac{125}{27} - \frac{25}{9} - \frac{25}{3} - 3\right) = -ve$$



154. Let the function $f(x) = \frac{x}{3} + \frac{3}{x} + 3$, $x \neq 0$ be strictly increasing in $(-\infty, \alpha_1) \cup (\alpha_2, \infty)$ and strictly decreasing in $(\alpha_3, \alpha_4) \cup (\alpha_5, \alpha_6)$. Then $\sum_{i=1}^5 \alpha_i^2$ is equal to :

माना फलन $f(x) = \frac{x}{3} + \frac{3}{x} + 3$, $x \neq 0$, $(-\infty, \alpha_1) \cup (\alpha_2, \infty)$ में निरंतर वर्धमान है और $(\alpha_3, \alpha_4) \cup (\alpha_5, \alpha_6)$ में निरंतर ह्रासमान

है। तो $\sum_{i=1}^5 \alpha_i^2$ बराबर है :

(A) 40

(B) 48

(C) 36

(D) 28

Ans. C

Sol. $f'(x) = -\frac{3}{x^2} + \frac{1}{3} = \frac{x^2 - 9}{3x^2} = \frac{(x+3)(x-3)}{3x^2}$

$$\begin{array}{ccccccc} + & , & - & , & - & , & + \\ -3 & & 0 & & 3 & & \end{array}$$

$$f(x) \uparrow \text{ in } x \in (-\infty, -3) \cup (3, \infty)$$

$$f(x) \downarrow \text{ in } x \in (-3, 0) \cup (0, 3)$$

$$\alpha_1 = -3, \alpha_2 = 3, \alpha_3 = -3, \alpha_4 = 0, \alpha_5 = 3$$

$$\sum_{i=1}^5 \alpha_i^2 = 9 + 9 + 9 + 0 + 9 = 36$$

155. Let $(2, 3)$ be the largest open interval in which the function $f(x) = 2\log_e(x-2) - x^2 + ax + 1$ is strictly increasing and (b, c) be the largest open interval, in which the function $g(x) = (x-1)^3(x+2-a)^2$ is strictly decreasing. Then $100(a+b-c)$ is equal to :

माना कि $(2, 3)$ वह सबसे बड़ा विवृत अंतराल है जिसमें फलन $f(x) = 2\log_e(x-2) - x^2 + ax + 1$ निरंतर वर्धमान है और (b, c) वह सबसे बड़ा विवृत अंतराल है जिसमें फलन $g(x) = (x-1)^3(x+2-a)^2$ निरंतर ह्रासमान है। तब $100(a+b-c)$ बराबर है:

- (A) 420 (B) 280 (C) 360 (D) 160

Ans. C

Sol. $f'(x) = \frac{2}{x-2} - 2x + a \geq 0$

$$f''(x) = \frac{-2}{(x-2)^2} - 2 < 0 \quad \Rightarrow f'(x) \text{ is Decreasing function}$$

$$\therefore f'(3) \geq 0 \quad \Rightarrow 2 - 6 + a \geq 0$$

$$a \geq 4$$

Put $a = 4 \quad \Rightarrow \quad g'(x) = 3(x-1)^2(x-2)^2 + 2(x-1)^3(x-2) \leq 0$

$$(x-1)^2(x-2)(5x-8) \leq 0$$

$$\begin{array}{ccccccc} + & & + & & - & & + \\ | & & | & & | & & | \\ 1 & & 8/5 & & 2 & & \end{array}$$

$$\downarrow \text{ing in } \left(\frac{8}{5}, 2 \right)$$

$$100(a+b+c) = 10 \left(4 + \frac{8}{5} - 2 \right) = 360$$

Note : If we take $a > 4$ (say $a = 5$), then

$$f'(x) = \frac{2}{x-2} - 2x + 5 \geq 0$$

$$\frac{2 - 2x^2 + 4x + 5x - 10}{x-2} \geq 0$$

$$\frac{2x^2 - 9x + 8}{x-2} \leq 0$$

$$\begin{array}{ccccccc} & & - & & + & & \\ | & & | & & | & & | \\ \frac{9-\sqrt{17}}{4} & & 2 & & \frac{9+\sqrt{17}}{4} & & \end{array}$$

$$f(x) \text{ is } \uparrow \text{ing in } \left(2, \frac{9+\sqrt{17}}{4} \right) \text{ which is greater than } (2, 3)$$

156. For the function $f(x) = \sin x + 3x - \frac{2}{\pi}(x^2 + x)$, where $x \in \left[0, \frac{\pi}{2} \right]$ consider the following two statements :

(I) f is increasing in $\left(0, \frac{\pi}{2}\right)$

(II) f' is decreasing in $\left(0, \frac{\pi}{2}\right)$

Between the above two statements :

(A) only (I) is true

(B) only (II) is true

(C) both (I) and (II) are true

(D) neither (I) nor (II) is true

फलन

$f(x) = \sin x + 3x - \frac{2}{\pi}(x^2 + x)$, जहाँ $x \in \left[0, \frac{\pi}{2}\right]$, के लिए, निम्न दो कथनों पर विचार करें।

(I) f अंतराल $\left(0, \frac{\pi}{2}\right)$ में वर्धमान है

(II) f' अंतराल $\left(0, \frac{\pi}{2}\right)$ में ह्रासमान है

उपरोक्त दो कथनों में से :

(A) केवल (I) सत्य है

(B) केवल (II) सत्य है

(C) (I) तथा (II) दोनों सत्य है

(D) न तो (I) और न ही (II) सत्य है

Ans. C

Sol. $f(x) = \sin x + 3x - \frac{2}{\pi}(x^2 + x)$, Where $x \in \left[0, \frac{\pi}{2}\right]$

$$f'(x) = \cos x + 3 - \frac{2}{\pi}(2x + 1)$$

$$= \cos x - \frac{4x}{\pi} - \frac{2}{\pi} + 3$$

$$\text{as } x \in \left[0, \frac{\pi}{2}\right]$$

$$\frac{4x}{\pi} \in [0, 2]$$

$$\Rightarrow 3 - \frac{2}{\pi} - \frac{4x}{\pi} > 0$$

and also $\cos x > 0$ when $x \in \left[0, \frac{\pi}{2}\right]$

$$\Rightarrow f'(x) > 0$$

$\Rightarrow F(x)$ is increasing

$$\text{Now, } f''(x) = -\sin x - \frac{4}{\pi} < 0 \forall x \in \left[0, \frac{\pi}{2}\right]$$

Hence, $f'(x)$ is decreasing

$\therefore$ Both statements (I) and (II) are true

157. The number of critical points of the function $f(x) = (x-2)^{\frac{2}{3}}(2x+1)$ is :

फलन $f(x) = (x-2)^{\frac{2}{3}}(2x+1)$ के क्रांतिक बिन्दुओं की संख्या होगी :

(A) 0

(B) 1

(C) 2

(D) 3

Ans. C

Sol. $f(x) = (x-2)^{\frac{2}{3}}(2x+1)$

$$f'(x) = 2(x-2)^{\frac{2}{3}} + (2x+1) \times \frac{2}{3}(x-2)^{-\frac{1}{3}}$$

$$f'(x) = 2(x-2)^{\frac{2}{3}} + \frac{2(2x+1)}{3(x-2)^{\frac{1}{3}}}$$

$$f'(x) = \frac{6(x-2) + 2(2x+1)}{3(x-2)^{\frac{1}{3}}}$$

$$f'(x) = \frac{10x-10}{3(x-2)^{\frac{1}{3}}} = \frac{10(x-1)}{3(x-2)^{\frac{1}{3}}}$$

Critical points 1,2

$\Rightarrow$ 2 critical points

158. For the function $f(x) = (\cos x) - x + 1$, $x \in \mathbb{R}$, between the following two statements :

(S1) $f(x) = 0$ for only one value of x in $[0, \pi]$.

(S2) $f(x)$ is decreasing in $\left[0, \frac{\pi}{2}\right]$ and increasing in $\left[\frac{\pi}{2}, \pi\right]$.

(A) Only (S2) is correct.

(B) Only (S1) is correct.

(C) Both (S1) and (S2) are correct.

(D) Both (S1) and (S2) are incorrect.

फलन $f(x) = (\cos x) - x + 1$, $x \in \mathbb{R}$ के लिए निचे दिये गये दो कथनों में से :

(S1) $[0, \pi]$ में x के केवल एक मान के लिए $f(x) = 0$ है

(S2) $\left[0, \frac{\pi}{2}\right]$ में $f(x)$ ह्रासमान और $\left[\frac{\pi}{2}, \pi\right]$ में वृद्धमान है

(A) केवल (S2) सत्य होगा

(B) केवल (S1) सत्य होगा

(C) (S1) तथा (S2) दोनों सत्य होंगे

(D) (S1) तथा (S2) दोनों असत्य होंगे

Ans. B

Sol. $f(x) = \cos x - x + 1$

$$f'(x) = -\sin x - 1$$

f is decreasing $\forall x \in \mathbb{R}$

$$f(x) = 0$$

$$f(0) = 2, f(\pi) = -\pi$$

f is strictly decreasing in $[0, \pi]$ and $f(0) \cdot f(\pi) < 0$

$\Rightarrow$ Only one position of $f(x) = 0$

S1 is correct and S2 is incorrect.

159. Consider the function $f : \left[\frac{1}{2}, 1\right] \rightarrow \mathbb{R}$ defined by $f(x) = 4\sqrt{2}x^3 - 3\sqrt{2}x - 1$. Consider the statements :

(I) The curve $y = f(x)$ intersects the x -axis exactly at one point

(II) The curve $y = f(x)$ intersects the x -axis at $x = \cos \frac{\pi}{12}$.

Then :

(A) Only (I) is correct

(B) Both (I) and (II) are the correct

(C) Both (I) and (II) are incorrect

(D) Only (II) is correct

$f(x) = 4\sqrt{2}x^3 - 3\sqrt{2}x - 1$ द्वारा परिभाषित फलन $f : \left[\frac{1}{2}, 1\right] \rightarrow \mathbb{R}$ के लिए कथनों :

(I) वक्र $y = f(x)$, x -अक्ष को मात्र एक बिन्दु पर काटता है।

(II) वक्र $y = f(x)$, x -अक्ष को $x = \cos \frac{\pi}{12}$ पर काटता है।

में से :

(A) केवल (I) सही है

(B) (I) तथा (II) दोनों सही हैं

(C) (I) तथा (II) दोनों गलत हैं

(D) केवल (II) सही हैं

Ans. B

Sol. Let $x = \frac{\pi}{12}$

$$\therefore 3x = \frac{\pi}{4}$$

$$\cos(3x) = \frac{1}{\sqrt{2}}$$

$$\Rightarrow 4\cos^3 x - 3\cos x = \frac{1}{\sqrt{2}}$$

$$\Rightarrow 4\sqrt{2}\cos^3 x - 3\sqrt{2}\cos x - 1 = 0$$

$\therefore x = \left(\frac{\pi}{12}\right)$ is the solution of above equation statement II is correct

$$f'(x) = 12\sqrt{2}x^2 - 3\sqrt{2}$$

Which is positive for $x \in \left[\frac{1}{2}, 1\right]$

Hence, only one solution in $\left[\frac{1}{2}, 1\right]$

$\therefore$ Statement I is also correct

160. The function $f(x) = \frac{x}{x^2 - 6x - 16}$, $x \in \mathbb{R} - \{-2, 8\} ::$

(A) decreases in $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$

(B) increases in $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$

(C) decreases in $(-2, 8)$ and increases in $(-\infty, -2) \cup (8, \infty)$

(D) decreases in $(-\infty, -2)$ and increases in $(8, \infty)$

फलन $f(x) = \frac{x}{x^2 - 6x - 16}$, $x \in \mathbb{R} - \{-2, 8\} ::$

(A) $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$ में ह्रासमान है

(B) $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$ में वर्धमान है

(C) $(-2, 8)$ में ह्रासमान है तथा $(-\infty, -2) \cup (8, \infty)$ वर्धमान है

(D) $(-\infty, -2)$ में ह्रासमान है तथा $(8, \infty)$ में वर्धमान है

Ans. A

Sol. $f(x) = \frac{x}{x^2 - 6x - 16}$

Now,

$$f'(x) = \frac{-(x^2 + 16)}{(x^2 - 6x - 16)^2}$$

$$f'(x) < 0$$

Thus $f(x)$ is decreasing in $(-\infty, -2) \cup (-2, 8) \cup (8, \infty)$